

File Types

Here are the major types of Tableau files that we work with in Desktop:

- **.twb** – A Tableau Workbook. This is an XML file containing the metadata for a Tableau workbook that includes information about the connections to the data source(s), what worksheets and dashboards are in the workbook, calculated fields created in Tableau, default color assignments, Tableau parameters, etc.
- **.tde** – A Tableau Data Extract that has the data from a data source stored in a columnar, highly compressed format.
- **.tds** – A Saved Data Source. This is an XML file that has the connection information, calculated fields, color assignments, etc. that we might find in a .twb, only stored as a separate file so it's reusable.
- **.twbx** – A Tableau Packaged Workbook. This is a zip file (with a different extension) that stores the .twb and all data sources (not all data sources can be exported into a .twbx, for example cubes).
- **.tdsx** – a zip file containing a .tds and the contents of the data source. I know this is possible, I've never seen one actually used in practice.

When we open up a .twb file in Tableau Desktop, Tableau reads the connection information and connects to the data source(s). When we open up a .twbx file, Tableau first unzips the .twbx (which includes the .twb and the data) into the user's temp folder (the location depends on the OS and the user's profile) and then works with that in place.

Data Extracts

When we're extracting data, there are a few different scenarios depending on whether the Tableau workbook is a .twb or .twbx.

1. If we have a .twb (or haven't saved a workbook yet), Tableau will prompt us to save the extract as a file.
2. For a .twbx, there are two outcomes:
 - If we use the "Import all data" / "Import some data" options when we make the connection, Tableau will prompt us to save the extract as a file.
 - If we connect live then in the Data window right-click on the data source and choose Extract Data..., the extract is saved **inside** the .twbx.

Whatever way we do things, the workbook (the XML) also includes the connection information back to the live data source so the extract can be refreshed.

Saving .twb vs. .twbx

When we save a .twb, the .twb has the connection info but not the data, so all the data lives separately. When we save a .twbx, if there's a live connection then Tableau will attempt to save that data source **inside** the .twbx (not all data sources can be saved inside a .twbx, for example cube sources), which effectively makes the data source static.

Classifying Data Connections

When it comes to data connections, I have several classifications that I keep in mind.

- **Live connection** to the data source.
- A **static data source** in a .twbx, created by saving a workbook with a live connection as a .twbx.
- An **extract that is made from a live connection to the data source**. This gets the double-cylinder icon.
- A **“naked” extract** – we can directly connect to a .tde file as a data source. This gets a single-cylinder icon and is a variation on the live connection, only it's really a static connection because the raw .tde doesn't have the information needed us to trigger a refresh. I don't really recommend this unless you are very sure of what you're doing, if the .tde changes that can case workbooks built on the naked extract to start throwing up errors, showing [Calculation_2987347315671] instead of your actual field names, etc. Use a .tds instead (a bit more about that below).
- A connection to a **Tableau Server Published Data Source** – this has four subtypes, the first are variations of three types mentioned above – a passthrough live connection to the data source, an extract made from a live connection, or a naked extract, and the fourth is a “static” connection to the data source (which is what we get from a live connection to a data source when we leave the “Include External Files” option to it's default of being checked, or make a connection to a static source where we Include External Files, there's a failure mode in this that I'll talk about further down in publishing).

An aside here: One way to think of .tds files is that they “wrap” any of the above connection types in metadata to make the connections re-usable, so instead of making the connection to the raw Excel file, SQL Server, Tableau Saved Data Source, etc. we “connect” to the .tds.

Some Wrinkles

There's a wrinkle in the above that some users run into:

1. Make a workbook
2. Connect live to a file-based data source, like an Excel file.
3. Save it as a .twbx. Tableau stores a copy of the data in the .twbx, thereby making it static.
4. Open up the .twbx.
5. Try to refresh the data, nothing happens. Grrr. Argh.

Here's another wrinkle:

1. Make a workbook
2. Connect live to a file-based data source, like an Excel file.
3. Save it as a .twbx. Tableau stores a copy of the data in the .twbx, thereby making it static.
4. Create an extract of the data. Tableau puts the extract **of the static data** in the .twbx.
5. Open up the .twbx.
6. Try to refresh the data, nothing happens. Grrr. Argh.

Here's a way that avoids either wrinkle:

1. Connect live to your data, whatever it is.
2. Extract the data.

3. **Then** save the .twbx.
4. Try to refresh the extract, and it will refresh. Yeay!

Publishing to Tableau Server

Ok, let's talk about Tableau Server now: Whether the workbook is a .twb or .twbx, Tableau always publishes the metadata (the XML file part as well as connection information) to Tableau Server. Whether the data is included depends on the kind of connection:

- When we publish a workbook to Tableau with a live connection to a non-file based data source such as SQL Server, Oracle, etc., then we can choose the Authentication method and go from there.
- When we publish a workbook to Tableau and the workbook has a live connection (including a connection to a naked extract) to a file-based data source, we get the "Include External Files" option. This default puts a **copy** of the live connection's data into Tableau Server (more on that in a minute), thereby making it a static data source. When we turn that option off, then Tableau Server won't copy the data and will attempt to use that live connection. Which is why we need to be careful about file paths—use UNC paths to be safe—and permissions, so the Tableau Server run-as user can also access your data. If Tableau Server can't connect, depending on whether the current worksheet is accessing that data source you'll get "An unexpected error occurred on the server." message, and possibly be sad and frustrated. You'll need to clear up the path & permissions problems.
- If the workbook has a static data source (as in publishing a static file that's inside a .twbx), we will also get the "Include External Files" option. If we leave the default on, that static data is copied to Tableau Server. If we uncheck that option, Tableau Server won't copy the data and will attempt to use a live connection back to the source, only the source only lives inside the temp file created on your hard drive when Tableau unzipped the .twbx so Tableau Server is almost certainly not going to be able to connect to it and you'll get that same "An unexpected error occurred on the server." message after publishing.
- When we publish a workbook to Tableau that uses extracts made from a live connection, Tableau copies the extract into Tableau Server, and preserves the connection information about how to get back to the live connection so the extract can be refreshed. Plus we get the Scheduling & Authentication option. (So even if you've used a .twb and stored the .tde separately, once you've published it Tableau Server ignores the .tde file, instead using and refreshing Tableau Server's own internal copy).
- A workbook using a Tableau Server Published Data Source connection is a special variation of a live connection that is always live and doesn't get the "Include External Files" option. (Even though the Tableau Server Published Data Source might be totally static).

If we're using multiple data sources in a workbook, we'll see an appropriate combination of the above options when publishing the workbook. Wee!

When we're creating a Tableau Saved Data Source, the process is the same as when we publish a .twb or .twbx.

Where art thou, data extract?

When I wrote "copies into Tableau Server", what happens is:

- The XML gets copied into Tableau's PostgreSQL database.
- File-based data sources that are copied due to "Include External Files" being on are copied into PostgreSQL.
- During the publishing process Tableau data extracts are copied into a Tableau server temp folder, and copied (and often renamed) into a more permanent location under ...Tableau Server\data\tabsvc\dataengine\folder_[#####]\upload_[GUID], where the the [#####] is an unknown-Tableau-assigned number (maybe something to do with versioning of installs?) and the GUID is a unique identifier of the Tableau workbook (I think). This appears to be true whether the extract is a naked extract (that can't be updated) or one that has the connection info and can be updated.
- I'm not sure what happens with custom shapes & background images. Since custom shapes are stored as [base64 in the XML](#), I'm guessing they end up in PostgreSQL as well. For .twb's background images are left in place as external files, for a .twbx they are stored as files inside the .twbx. I haven't been able to find the background images anywhere on Tableau Server except in a temp folder so I suspect they go into PostgreSQL. If you have more information on this, please let me know.

Note that while a Tableau Server user is uploading and/or interacting with a workbook we will find various temp files with data on Tableau Server, but those get blown away as soon as the interaction is over and Tableau Server does its cleanup.

It's possible to update the .tde file in Tableau Server's ...dataengine\... folder structure on your own. And [some people](#) (I'm looking at you, [Tamar Foldi](#)) have gone into the PostgreSQL tables to do hacky-things, and all of that is way into undocumented/break your warranty territory.

Correction 2014-11-02: This section has been changed from the original post based on a correction from [Craig Bloodworth](#) (in the comments below) and further investigation. I also added the extra info on custom shapes & background images.

More (Tableau Server-flavored) Wrinkles

Here's a scenario, that is very close to a real-life situation that was one of my early experiences with Tableau Server when I was trying to figure this out.

1. Create a Tableau workbook.
2. Connect to a file-based data source, like an Excel file.
3. Save it as a .twbx. Tableau stores a copy of the data in the .twbx, thereby making it static.
4. Create an extract of the data. Tableau puts the extract **of the static data** in the .twbx.
5. Build your viz.
6. Publish the viz to Tableau Server, including setting an extract refresh schedule.
7. [time passes]
8. The extract refresh works perfectly, only refreshing static data.
9. [time passes]
10. The original data source gets updated with new records.
11. [time passes]
12. The extract refresh again works perfectly, only it never sees the new records, only the static ones.
13. User tries to see the updated data on Tableau, and doesn't.
14. User emails you saying "Where's my data?"
15. You scratch your head.
16. You download the workbook, try refreshing the data, turning off the extract, deleting the extract, etc., and eventually reconnect to the data.

17. But you repeat step #3 and have to go through the same thing over again. **GRR. ARGH.**
18. After getting bruises on your forehead from slamming your head against your desk, you fire up your search engine, and hopefully find this post that reminds you to be careful of what order you do things in.

Or, here's another scenario with a Tableau Server Published Data Source:

1. Make a workbook
2. Connect live to a file-based data source, like an Excel file.
3. Publish the data source to Tableau Server, leaving the default "Include External Files" checked. This makes the live connection now static, since Tableau **copies** the data to Tableau Server.
4. Reconnect to the Tableau Server Published Data Source.
5. Build the viz.
6. Publish the workbook to Tableau Server.
7. [time passes]
8. The original data source gets updated with new records.
9. User tries to see the updated data on Tableau, and doesn't.
10. User emails you saying "Where's my data?"
11. You scratch your head, maybe go back to step 2 and repeat, and life is a world of pain and suffering until you realize that you need to turn **off** Include External Files